

**VISVESVARAYA TECHNOLOGICAL UNIVERSITY,  
JNANASANGAMA, BELAGAVI – 590018**



**AMini Project Report On**

**“Online Voting System”**

*Submitted in partial fulfillment for the award of degree of*

**Bachelor of Engineering**

**In**

**Information Science & Engineering**

*Submitted by*

**DEEPAK D R**

**USN:4MO21IS014**

**Under The Guidance Of**

**Mr. Jayaram CV**

Asst.Prof.

Department of CSE

**MYCEM College.Mysore**

**Department of Computer Science & Engineering**

**MYSORE COLLEGE OF ENGINEERING & MANAGEMENT, MYSURU**

(Affiliated to VTU, Belagavi, Approved by AICTE, New Delhi & Govt. of Karnataka)

#072, T Narsipura Road, Near Big Banyan Tree Chikkahalli, Mysore, Karnataka -28

**DEPARTMENT OF COMPUTER SCIENCE**  
**MYSORE COLLEGE OF ENGINEERING AND MANAGEMENT**  
**MYSURU**

(Affiliated to VTU, Belagavi, Approved by AICTE, New Delhi & Govt. of Karnataka)  
#072, T Narsipura Road, Near Big Banyan Tree Chikkahalli, Mysore, Karnataka -28  
2023-2024



**CERTIFICATE**

This is to certify that the Mini Project work entitled ***“Online Voting System”*** is a presentation done by **Mr. DEEPAK D R (4MO21IS014)**, in partial fulfillment for the award of **Bachelor of Engineering** in **INFORMATION SCIENCE & ENGINEERING** of the Visvesvaraya Technological University, Belagavi, during the year 2023-2024. The Mini Project report has been approved as it satisfies the academic requirements in respect of Mini Project work prescribed for the bachelor of engineering

.....

**MR. Jayaram C V**

Asst.Prof.  
Department of CSE  
**MYCEM College. Mysore**

.....  
**Prof.HARSHAKUMAR H S**

HOD  
Dept. of CSE  
**MyCEM College. Mysore**

.....  
**Dr.M.S.PRABHUSWAMY**

PRINCIPAL  
**MyCEM College. Mysore**

**External Viva:**

**Examiner Name**

1).....  
2).....

**Signature with Date**

.....  
.....

## DECLARATION

We are extremely thankful to **Dr. MS Prabhuswamy., Principal, Mycem., Mysore**, for providing us the academic ambience and everlasting motivation to carry out this work and shaping our careers.

We express our sincere gratitude to **Mr. Harsha Kumar HS, HOD, Dept. of Information Science and Engineering, Mycem., Mysore**, for his stimulating guidance, continuous encouragement and motivation throughout the course of present work.

I extend my sincere appreciation to my guide **Mr. Jayaram CV, Assistant professor, Department Of Computer Science & Engineering, Mycem** who provided her valuable suggestions and precious time in accomplishing my midterm min project report.

We would also like to thank all other teaching and technical staffs of Department of Information Science and Engineering, who have directly or indirectly helped us in the completion of this Project Work.

Lastly, I would like to thank the almighty and my parents for their moral support and my friends with whom I shared my day-to-day experience and received lots of suggestions that improved my quality of work.

**DEEPAK DR (4MO21IS014)**

# ABSTRACT

The **android voting system** is an android application that provides a new technique of casting votes using mobile phones. This application is especially developed for organizations, corporations and commercial businesses to get employees opinions whenever there is any new policy implemented or any issue being investigated or during controversies. The project provides an effective solution in resolving all the arguments that occurs in organizations by considering all employees opinions.

In the traditional system there was a need to go on the voting booth and cast a vote. People from distinct places who did not have their voting cards cannot cast their votes. Also authentication of the user was not good and appropriate. There was a lot of paper work which was very time consuming. The results needed to be calculated manually which was very time consuming process. Therefore the proposed system is developed to remove the efforts needed in the traditional voting process.

The proposed system has an application developed on android phone via which the user can cast his vote from anywhere on the face of the globe. The user registers by giving his personal details which gets stored in the database at the server side.

# CONTENTS

|                                        | PAGE NO |
|----------------------------------------|---------|
| <b>1. CHAPTER 1</b>                    |         |
| INTRODUCTION                           | 1       |
| 1.1 EXISTING SYSTEM.....               | 1       |
| 1.2 MOTIVATION .....                   | 1       |
| 1.3 PROBLEM STATEMENT .....            | 2       |
| 1.4 OBJECTIVE OF PROJECT .....         | 2       |
| 1.5 PROPOSED SYSTEM .....              | 2       |
| 1.6 SCOPE OF STUDY .....               | 3       |
| <b>2. CHAPTER 2</b>                    |         |
| LITERATURE SURVEY                      | 4       |
| 2.1 INTRODUCTION .....                 | 4       |
| 2.2 LITERATURE SURVEY .....            | 4       |
| 2.3 REVIEW CONCEPT.....                | 4       |
| 2.4 SURVEY PAPER.....                  | 5       |
| <b>3. CHAPTER 3</b>                    |         |
| SYSTEM REQUIREMENT SPECIFICATION       | 6       |
| 3.1 ANDROID OPERATING SYSTEM.....      | 6       |
| 3.2 FUNCTIONAL REQUIREMENTS.....       | 8       |
| 3.3 NON-FUNCTIONAL REQUIREMENTS.....   | 8       |
| 3.3.1 ORGANIZATIONAL REQUIREMENTS..... | 9       |
| 3.3.2 USER REQUIREMENTS.....           | 9       |
| 3.3.3 BASIC OPERATIONAL.....           | 10      |
| 3.4 HARDWARE REQUIREMENTS.....         | 11      |
| 3.5 SOFTWARE REQUIREMENTS.....         | 11      |
| <b>4. CHAPTER 4</b>                    |         |
| METHODOLOGY                            | 12      |
| 4.1 INTRODUCTION.....                  | 12      |

|           |                                      |           |
|-----------|--------------------------------------|-----------|
| 4.2       | SYSTEM DEVELOPMENT METHODOLOGY.....  | 12        |
| 4.2.1     | MODEL PHASES.....                    | 12        |
| 4.2.2     | WATERFALL MODEL.....                 | 13        |
| 4.3       | DESIGN USING UML.....                | 14        |
| 4.4       | DATA FLOW DIAGRAM.....               | 14        |
| 4.5       | .CLASS DIAGRAM.....                  | 15        |
| 4.6       | USE CASE DIAGRAM.....                | 16        |
| 4.7       | ACTIVITY DIAGRAM.....                | 17        |
| 4.8       | SEQUENCE DIAGRAM.....                | 18        |
| <b>5.</b> | <b>CHAPTER 5</b>                     |           |
|           | IMPLEMENTATION                       | 19        |
| 5.1       | INTRODUCTION.....                    | 19        |
| 5.2       | INTERFACING ANDROID WITH SERVER..... | 19        |
| 5.3       | CHALLENGES AND MITIGATIONS .....     | 20        |
| <b>6.</b> | <b>CHAPTER 6</b>                     |           |
| 6.1       | INTRODUCTION.....                    | 25        |
| 6.2       | TESTING METHODOLOGIES.....           | 25        |
| 6.3       | TEST CASE 1.....                     | 27        |
| 6.4       | TEST CASE 2.....                     | 27        |
| 6.5       | TEST CASE 3.....                     | 27        |
| 6.6       | TEST CASE 4.....                     | 27        |
| 6.7       | TEST CASE 5.....                     | 28        |
| <b>7.</b> | <b>CHAPTER 7</b>                     |           |
| 7.1       | Conclusion.....                      | 33        |
| 7.2       | Future Scope.....                    | 33        |
| 7.3       | Research Issues.....                 | 34        |
| 7.4       | Implementation Issues.....           | 34        |
|           | <b>SNAPSHOT.....</b>                 | <b>29</b> |
|           | <b>REFERENCES.....</b>               | <b>35</b> |

## LIST OF FIGURE

|                                               | PAGE NO |
|-----------------------------------------------|---------|
| FIG 4.1 WATERFALL MODEL                       | 14      |
| FIG 4.2 CLASS DIAGRAM OF SYSTEM               | 15      |
| FIG 4.3 USE CASE DIAGRAM                      | 16      |
| FIG 4.4 ACTIVITY DIAGRAM                      | 17      |
| FIG 4.5 SEQUENCE DIAGRAM                      | 18      |
| SNAPSHOT 1 : APP ACTIVITIES-1(USER ACTIVITY)  | 30      |
| SNAPSHOT 2 : APP ACTIVITIES-1(ADMIN ACTIVITY) | 32      |

## CHAPTER 1

# INTRODUCTION

ANDROID BASED ONLINE VOTING SYSTEM is an online voting technique. In this system people who have citizenship of India and whose age is above 18 years of age and any sex can give his vote online without going to any physical polling station. There is a database which is maintained in which all the names of voters with complete information is stored.

In ANDROID BASED ONLINE VOTING SYSTEM a voter can use his voting right online without any difficulty. He has to be registered first for him to vote. Registration is mainly done by the system administrator for security reasons. The system Administrator registers the voters on a special site of the system visited by him only by simply filling a registration form to register voter. Citizens seeking registration are expected to contact the system administrator to submit their details. After the validity of them being citizens of India has been confirmed by the system administrator by comparing their details submitted with those in existing databases such as those as the Registrar of Persons, the citizen is then registered as a voter.

After registration, the voter is assigned a secret Voter ID with which he/she can use to log into the system and enjoy services provided by the system such as voting. If invalid/wrong details are submitted, then the citizen is not registered to vote.

### 1.1 Existing System

In existing system of voting, voter go on voting booth on the day of voting. There is no any centralized system, where we can cast our vote from remote location. There is no proper authentication present of an individual. There is manual work involved, so it may generate errors in the system. The Traditional approach is very time consuming and have got manual errors.

### 1.2 Motivation

Elections play a major role in the country. Usually elections follow the Traditional Paper based Voting or EVMs . Citizens has to move to the polling stations to cast their vote. It leads to gathering of people in large number .In this pandemic situation its not safe to encourage gatherings. And also we cannot avoid the elections. People at remote areas from their native places find difficult to reach their hometowns to cast vote due to various reasons . Aged people cannot travel to polling stations and wait for longer time in queue to cast vote. In order t overcome all these issues ,we decided to find a solution and that is on-line voting system.



### 1.3 Problem Statement

- **Expensive and Time consuming:** The process of collecting data and entering this data into the database takes too much time and is expensive to conduct, for example, time and money is spent in printing data capture forms, in preparing registration stations together with human resources, and there after advertising the days set for registration process including sensitizing voters on the need for registration, as well as time spent on entering this data to the database.
- **Too much paper work:** The process involves too much paper work and paper storage which is difficult as papers become bulky with the population size.
- **Errors during data entry:** Errors are part of all human beings; it is very unlikely for humans to be 100 percent efficient in data entry.
- **Loss of registration forms:** Some times, registration forms get lost after being filled in with voters details, in most cases these are difficult to follow-up and therefore many remain unregistered even though they are voting age nationals and interested in exercising their right to vote.
- **Short time provided to view the voter register:** This is a very big problem since not all people have free time during the given short period of time to check and update the voter register.
- Above all, a number of voters end up being locked out from voting.

### 1.4 Objective of Project

The specific objectives of the project include:

- Reviewing the existing/current voting process or approach in India.
- Coming up with an automated voting system in India.
- Implementing a an automated/online voting system.
- Validating the system to ensure that only legible voters are allowed to vote.

### 1.5 Proposed System

This system was proposed to eliminate the trouble of people to go and vote at the voting booth. Whenever schedule date notification is get on user android device, user can cast their vote from anywhere and at anytime. For casting the vote particular user should be authorized so the proposed system will done authentication of voter. With the help of mobile finger print device.

This proposed system is based on fingerprint authorization. During registration the proposed system will get all user details. And while user is signing in to the application this will authorize the user with the help of fingerprint device. In proposed system advancement in android device is very easy and secure voting. The proposed system provides the specification and requirements for E-Voting using an Android platform.

## **1.6 Scope of Study**

It is focused on studying the existing system of voting in India and to make sure that the peoples vote is counts, for fairness in the elective positions. This is also will produce:

- Less effort and less labor intensive, as the primary cost and focus primary on creating, managing, and running a secure web voting portal.
- Increasing number of voters as individuals will find it easier and more convenient to vote, especially those abroad.

## CHAPTER 2

# LITERATURE SURVEY

### 2.1 Introduction

Literature survey is mainly carried out in order to analyze the background of the current project which helps to find out flaws in the existing system and guides on which unsolved problems we can work out. So, the following topics not only illustrate the background of the project but also uncover the problems and flaws which motivated to propose solutions and work on this project.

### 2.2 Literature Survey

To make the voting process very easy and efficient wireless and web technologies are used. The online-voting system has the possibility of secure, easy and safe way to capture and count the votes in the election. The proposed system provides the specification and requirements for Online-Voting using an Android platform. The Online-voting means the voting process in election by using mobile phone. The android platform is used to develop an Online-voting application. Through a general diagram the introduction of the system is presented. The proposed Online-voting system will be presented with the obtained results. The proposed system also described how the android mobile phones are efficient. The android platform is used to develop a reliable and efficient application. Using the face-book APIs provided by the android SDK (software development kit) the login can be done very efficiently.

### 2.3 Review Concept

- **Efficient E-voting Android Based System:** The advancement in the mobile devices the wireless sensor network and web technologies that may given rise to the new applications which will make the voting process very easy and efficient. The research project provides the specifications and requirements for the E-Voting using an Android Platform. The E-Voting means the voting process in election by using electronic device. E-Voting is not more secure as possible than the paper-ballot system. Electronic failures might occur with such a system.
- **Web-Based Voting System Using Fingerprint:** The problem of voting system is still critical in terms of safety and the security. The design and development of the web based voting system using fingerprint verification in order to provide a high performance with high security to the voting system also the user can use the web technology to make the voting system more practical.

The proposed architecture presents the module for election system for selecting the president in university. The Electronic voting System matched with an already saved image with in a database. Web based Voting System using Fingerprint Recognition. The online system has provided an efficient way to cast their votes, free of fraud, and make the system more trust, economic and fast. More number of peoples were participate in the election to vote their candidate for choosing the majority votes. When limited number of people were register in the system means that particular candidate only allowed for voting and the remaining peoples may not allowed for the voting.

## 2.4 Survey Papers

[1] “Online Voting System” uses fingerprint and uses Aadhaar card for authentication. This system is very efficient and ads extra security to avoid occurrence of fraud. In paper

[2] “Online voting system” we collected most the part of our system from this paper login id and password is used for authentication purpose and admin panel controls and look after each and every aspect from registration to the result creation. In paper

[3] “Online voting system using Cloud” UI consists of login name and some special characters for user login purpose. Register voter has to use his/her email id for creation for unique id which is essential at time of login.

- **Title:** Election voting system using mobile (m-voting)-(2013)

**Context:** Biometric identification of voter by using iris pattern scanning and OTP of each individual voter for validation and authorization purpose.

- **Title:** Event based application of voting system for mobile devices-(2015)

**Context:** An objective was to focus on implementing mobile voting application for measuring the cognitive competencies of social group, particularly in the field of education.

- **Title:** Mobile based facial recognition using OTP verification for voting system -(2015)

**Context:** Biometric identification of voter by using face recognition and OTP of each individual voter for validation and authorization purpose.

## CHAPTER 3

# SYSTEM REQUIREMENT SPECIFICATION

### 3.1 Android Operating System

Android is a mobile operating system developed by Google, based on a modified version of the Linux kernel and other open source software and designed primarily for touchscreen mobile devices such as smartphones and tablets. In addition, Google has further developed Android TV for televisions, Android Auto for cars, and Wear OS for wrist watches, each with a specialized user interface. Variants of Android are also used on game consoles, digital cameras, PCs and other electronics

Initially developed by Android Inc., which Google bought in 2005, Android was unveiled in 2007, with the first commercial Android device launched in September 2008. The operating system has since gone through multiple major releases, with the current version being 8.1 "Oreo", released in December 2017. The core Android source code is known as Android Open Source Project (AOSP), and is primarily licensed under the Apache License

Android is also associated with a suite of proprietary software developed by Google, including core apps for services such as Gmail and Google Search, as well as the application store and digital distribution platform Google Play, and associated development platform. These apps are licensed by manufacturers of Android devices certified under standards imposed by Google, but AOSP has been used as the basis of competing Android ecosystems, such as Amazon Fire OS, which utilize its own equivalents to these Google Mobile Services.

Android has been the best-selling OS worldwide on smartphones since 2011 and on tablets since 2013. As of May 2017, it has over two billion monthly active users, the largest installed base of any operating system, and as of 2017, the Google Play store features over 3.5 million apps.

- **Interface** Android's default user interface is mainly based on direct manipulation using touch inputs that loosely correspond to real-world actions, like swiping, tapping, pinching, and reverse pinching to manipulate on-screen objects, along with a virtual keyboard. Game controllers and full-size physical keyboards are supported via Bluetooth or USB. The response to user input is designed to be immediate and provides a fluid touch interface, often using the vibration capabilities of the device to provide haptic feedback to the user

- **Applications** Application ("apps"), which extend the functionality of devices, are written using the Android software development kit (SDK) and, often, the Java programming language. Java may be combined with C/C++, together with a choice of non-default runtimes that allow better C++ support. The Go programming language is also supported, although with a limited set of application programming interfaces (API). In May 2017, Google announced support for Android app development in the Kotlin programming language.
- **Memory Management** Since Android devices are usually battery-powered, Android is designed to manage processes to keep power consumption at a minimum. When an application is not in use the system suspends its operation so that, while available for immediate use rather than closed, it does not use battery power or CPU resources. Android manages the applications stored in memory automatically: when memory is low, the system will begin invisibly and automatically closing inactive processes, starting with those that have been inactive for the longest amount of time. Lifehacker reported in 2011 that thirdparty task killer applications were doing more harm than good.
- **Hardware** The main hardware platform for Android is ARM (the ARMv7 and ARMv8-A architectures), with x86, MIPS and MIPS64, and x86-64 architectures also officially supported in later versions of Android. The unofficial Android-x86 project provided support for x86 architectures ahead of the official support. The MIPS architecture was also supported ahead of Google's support. Since 2012, Android devices with Intel processors began to appear, including phones and tablets. While gaining support for 64-bit platforms, Android was first made to run on 64-bit x86 and then on ARM64. Since Android 5.0 "Lollipop", 64-bit variants of all platforms are supported in addition to the 32-bit variants.

Requirements for the minimum amount of RAM for devices running Android 7.1 range from in practice 2 GB for best hardware, down to 1 GB for the most common screen, to absolute minimum 512 MB for the lowest spec 32-bit smartphone. The recommendation for Android 4.4 is to have at least 512 MB of RAM while for "low RAM" devices 340 MB is the required minimum amount that does not include memory dedicated to various hardware components such as the baseband processor. Android 4.4 requires a 32-bit ARMv7, MIPS or x86 architecture processor (latter two through unofficial ports), together with an OpenGL ES 2.0 compatible graphics processing unit (GPU). Android supports OpenGL ES 1.1, 2.0, 3.0, 3.1 and as of latest major version, 3.2 and since Android 7.0 Vulkan (and version 1.1 available for some devices

## 3.2 Functional Requirements

Functional Requirement defines a function of a software system and how the system must behave when presented with specific inputs or conditions. These may include calculations, data manipulation and processing and other specific functionality. In this system following are the functional requirements:-

- Input test case must not have compilation and runtime errors.
- The application must not stop working when kept running for even a long time.
- The application must function as expected for every set of test cases provided.
- The application should generate the output for given input test case and input parameters.
- The application should generate on-demand services.

## 3.3 Non-Functional Requirements

Non-functional requirements are the requirements which are not directly concerned with the specific function delivered by the system. They specify the criteria that can be used to judge the operation of a system rather than specific behaviors. They may relate to emergent system properties such as reliability, response time and store occupancy. Non-functional requirements arise through the user needs, because of budget constraints, organizational policies, the need for interoperability with other software and hardware systems or because of external factors such as:-

- Product Requirements
- Organizational Requirements
- User Requirements
- Basic Operational Requirements

In systems engineering and requirements engineering, a non-functional requirement is a requirement that specifies criteria that can be used to judge the operation of a system, rather than specific behavior . This should be contrasted with functional requirements that define specific behavior or functions. The plan for implementing non-functional requirements is detailed in the system architecture. Broadly, functional requirements define what a system is supposed to do and non- functional requirements define how a system is supposed to be.

form of system shall do ;requirement¿, an individual action of part of the system, perhaps explicitly in the sense of a mathematical function, a black box description input, output, process and control functional

model or IPO Model. In contrast, nonfunctional requirements are in the form of system shall be requirement, an overall property of the system as a whole or of a particular aspect and not a specific function. The systems' overall properties commonly mark the difference between whether the development project has succeeded or failed.

### 3.3.1 Product Requirements

- **Portability:** Since the system is designed to run using Android, the system is portable.
- **Correctness:** It follows a well-defined set of procedures and rules to compute and also rigorous testing is performed to confirm the correctness of the data.
- **Ease of Use:** The front end is designed in such a way that it provides an interface which allows the user to interact in an easy manner.
- **Modularity:** The complete product is broken up into many modules and welldefined interfaces are developed to explore the benefit of flexibility of the product.
- **Robustness:** This software is being developed in such a way that the overall performance is optimized and the user can expect the results within a limited time with utmost relevancy and correctness .whereas evolution quality involves testability, maintainability, extensibility or scalability.

#### 3.3.1.1 Organizational Requirements Process Standards:

IEEE standards are used to develop the application which is the standard used by the most of the standard software developers all over the world. This stage is the first step in moving from problem to the solution domain. In other words, starting with what is needed design takes us to work how to satisfy the needs

#### 3.3.1.2 User Requirements

The user requirements document (URD) or user requirements specification is a document usually used to software engineering that specifies the requirements user expects from software to be constructed in software project. Once the required information is completely gathered it is documented in a URD, which is meant to spell out exactly

what the software must do and becomes part of the contractual agreement. A customer cannot demand feature not in the URD, whilst the developer cannot claim the product is ready if it does not meet an item of the URD. The URD can be used as a guide to planning cost, timetables, milestones, testing etc. The explicit nature of the URD allows customers to show it to various stakeholders to make sure all necessary



features are described. Formulating a URD requires negotiation to determine what is technically and economically feasible. Preparing a URD is one of those skills that lies between a science and economically feasible. Preparing a URD is one of those skills that lies between a science and an art, requiring both software technical skills and interpersonal skills.

### 3.3.1.3 Basic Operational Requirements

Operational requirement is the process of linking strategic goals and objectives to tactic goals and objectives. It describes milestones, conditions for success and explains how, or what portion of, a strategic plan will be put into operation during a given operational period, in the case of, a strategic plan will be put into operation during a given operational period, in the case of commercial application, a fiscal year or another given budgetary term. An operational plan is the basis for, and justification of an annual operating budget request. Therefore, a five-year strategic plan would typically require five operational plans funded by five operating budgets. Operational plans should establish the activities and budgets for each part of the organization for the next 1-3 years. They link the strategic plan with the activities the organization will deliver and the resources required to deliver them. An operational plan draws directly from agency and program strategic plans to describe agency and program missions and goals, program objectives, and program activities. Like a strategic plan, an operational plan addresses four questions:

- Where are we now?
- Where do we want to be?
- How do we get there?

The customers are those that perform the eight primary functions of systems engineering, with special emphasis on the operator as the key customer. Operational requirements will define the basic need and, at a minimum, will be related to these following points:

- **Mission profile or scenario:** It describes about the procedures used to accomplish mission objective. It also finds out the effectiveness or efficiency of the system.
- **Performance and related parameters:** It points out the critical system parameters to accomplish the mission
- **Utilization environments:** It gives a brief outline of system usage. Finds out appropriate environments for effective system operation.
- **Operational life cycle:** It defines the system lifetime

### 3.4 Hardware Requirements

- Processor: 800MHz Intel Pentium III or equivalent • Memory: 512 MB
- Disk space: 750 MB of free disk space

For Mobile:

- Running Device : Android.
- RAM : 512 MB minimum.
- Internal storage :1 GB.
- Internet access : Yes.

### 4.5 Software Requirements

- Operating System : Windows 10 and Ubuntu.
- Coding Language : Java.
- Tools : Eclipse and IDE.
- Library support : SDK,JDK.

## CHAPTER 4

# METHODOLOGY

### 4.1 Introduction

Design is a meaningful engineering representation of something that is to be built. It is the most crucial phase in the developments of a system. Software design is a process through which the requirements are translated into a representation of software. Design is a place where design is fostered in software Engineering. Based on the user requirements and the detailed analysis of the existing system, the new system must be designed. This is the phase of system designing. Design is the perfect way to accurately translate a customer requirement in the finished software product. Design creates a representation or model, provides details about software data structure, architecture, interfaces and components that are necessary to implement a system. The logical system design arrived at as a result of systems analysis is converted into physical system design.

### 4.2 System Development Methodology

System development method is a process through which a product will get completed or a product gets rid from any problem. Software development process is described as a number of phases, procedures and steps that gives the complete software. It follows series of steps which is used for product progress. The development method followed in this project is waterfall model.

#### 4.2.1 Model Phases

The waterfall model is a sequential software development process, in which progress is seen as flowing steadily downwards (like a waterfall) through the phases of Requirement initiation, Analysis, Design, Implementation, Testing and maintenance.

- **Requirement Analysis:** This phase is concerned about collection of requirement of the system. This process involves generating document and requirement review.
- **System Design:** Keeping the requirements in mind the system specifications are translated in to a software representation. In this phase the designer emphasizes on:-algorithm, data structure, software architecture etc.
- **Coding:** In this phase programmer starts his coding in order to give a full sketch of product. In other words system specifications are only converted in to machine readable compute code.

- **Implementation:** The implementation phase involves the actual coding or programming of the software. The output of this phase is typically the library, executables, user manuals and additional software documentation
- **Testing:** In this phase all programs (models) are integrated and tested to ensure that the complete system meets the software requirements. The testing is concerned with verification and validation.
- **Maintenance:** The maintenance phase is the longest phase in which the software is updated to fulfill the changing customer need, adapt to accommodate change in the external environment, correct errors and oversights previously undetected in the testing phase, enhance the efficiency of the software.

#### 4.2.2 Reason For Choosing Waterfall Model As Development Method

- Clear project objectives.
- Stable project requirements.
- Progress of system is measurable.
- Strict sign-off requirements.
- Helps you to be perfect.
- Logic of software development is clearly understood.
- Production of a formal specification
- Better resource allocation.
- Improves quality. The emphasis on requirements and design before writing a single line of code ensures minimal wastage of time and effort and reduces the risk of schedule slippage.
- Less human resources required as once one phase is finished those people can start working on to the next phase.

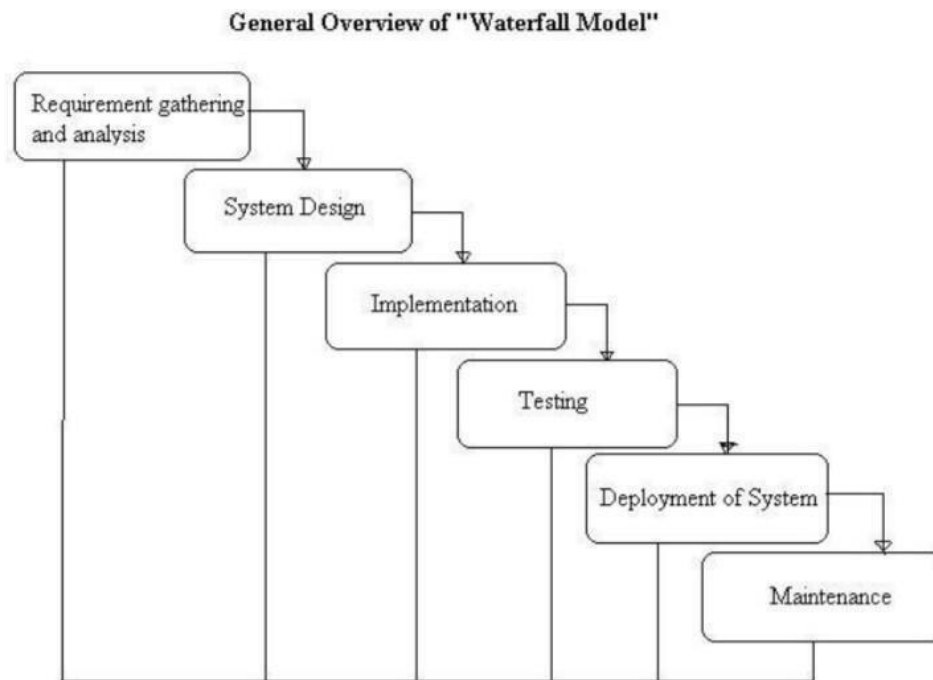


Fig 4.1: Waterfall Model

### 4.3 Design Using UML

Designing UML diagram specifies, how the process within the system communicates along with how the objects within the process collaborate using both static as well as dynamic UML diagrams since in this ever-changing world of Object Oriented application development, it has been getting harder and harder to develop and manage high quality applications in reasonable amount of time. As a result of this challenge and the need for a universal object modeling language every one could use, the Unified Modeling Language (UML) is the Information industries version of blue print. It is a method for describing the systems architecture in detail. Easier to build or maintains system, and to ensure that the system will hold up to the requirement changes.

### 4.4 Data Flow Diagram

A data flow diagram (DFD) is graphic representation of the "flow" of data through an information system. A data flow diagram can also be used for the visualization of data processing (structured design). It is common practice for a designer to draw a context level DFD first which shows the interaction between

the system and outside entities. DFDs show the flow of data from external entities into the system, how the data moves from one process to another, as well as its logical storage.

There are only four symbols: 1. Squares representing external entities, which are sources and destinations of information entering and leaving the system. 2. Rounded rectangles representing processes, in other methodologies, may be called 'Activities', 'Actions', 'Procedures', 'Subsystems' etc. which take data as input, do processing to it, and output it. 3. Arrows representing the data flows, which can either, be electronic data or physical items. It is impossible for data to flow from data store to data store except via a process, and external entities are not allowed to access data stores directly. 4. The flat three-sided rectangle is representing data stores should both receive information for storing and provide it for further processing.

## 4.5 Class Diagram

UML class diagram shows the static structure of the model. The class diagram is a collection of static modeling elements, such as classes and their relationships, connected as a graph to each other and to their contents. The class diagram is the main building block of object oriented modeling. It is used both for general conceptual modeling of the systematic of the application, and for detailed modeling translating the models into programming code. Class diagrams can also be used for data modeling. The classes in a class diagram represent both the main objects and or interactions in the application and the objects to be programmed.

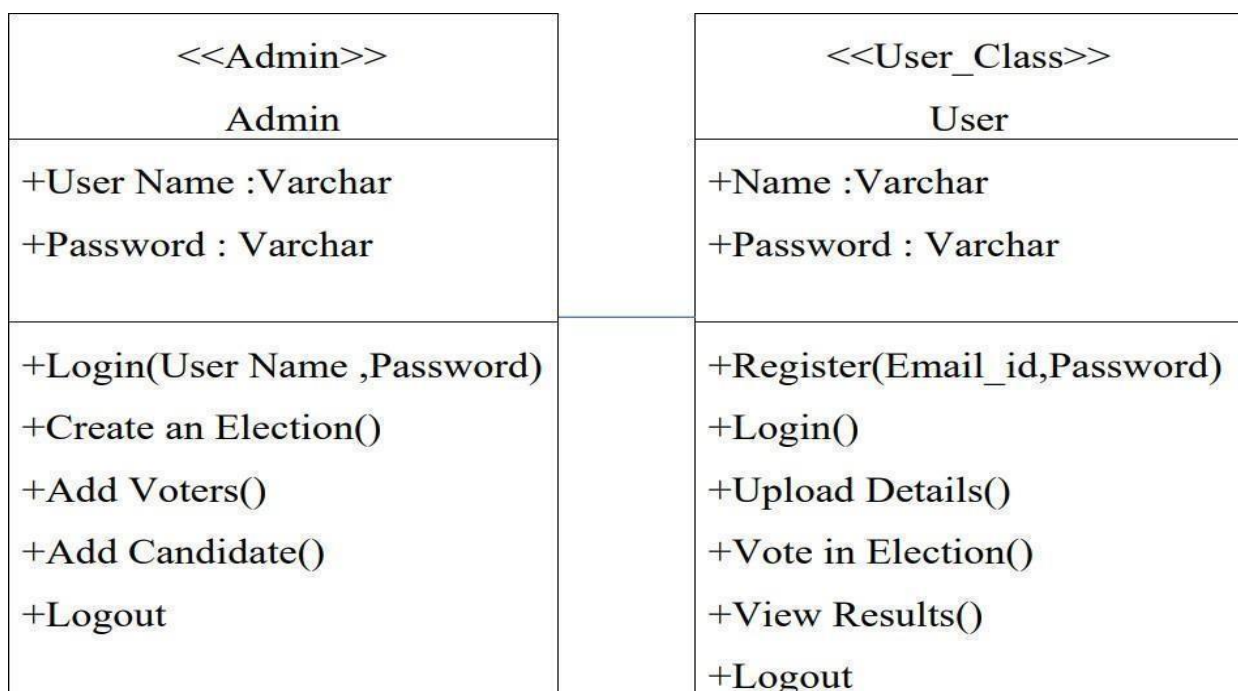


Fig 4.2: Class Diagram Of System

## 4.6 Use Case Diagram

A use case defines a goal-oriented set of interactions between external entities and the system under consideration. The external entities which interact with the system are its actors. A set of use cases describe the complete functionality of the system at a particular level of detail and it can be graphically denoted by the use case diagram.

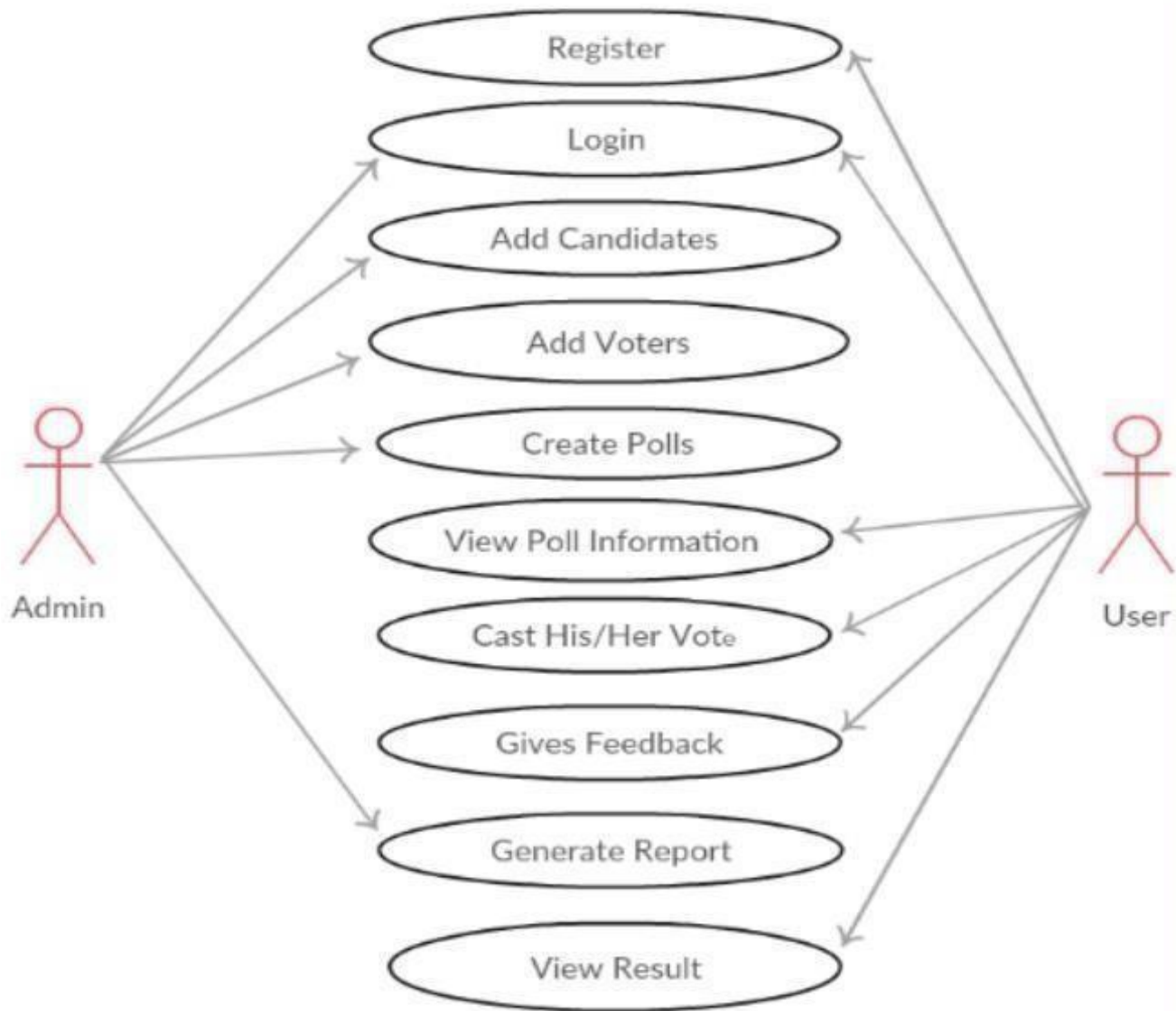


Fig 4.3: Use Case Diagram

## 4.7 Activity Diagram

An activity diagram shows the sequence of steps that make up a complex process. An activity is shown as a round box containing the name of the operation. An outgoing solid arrow attached to the end of the activity symbol indicates a transition triggered by the completion.

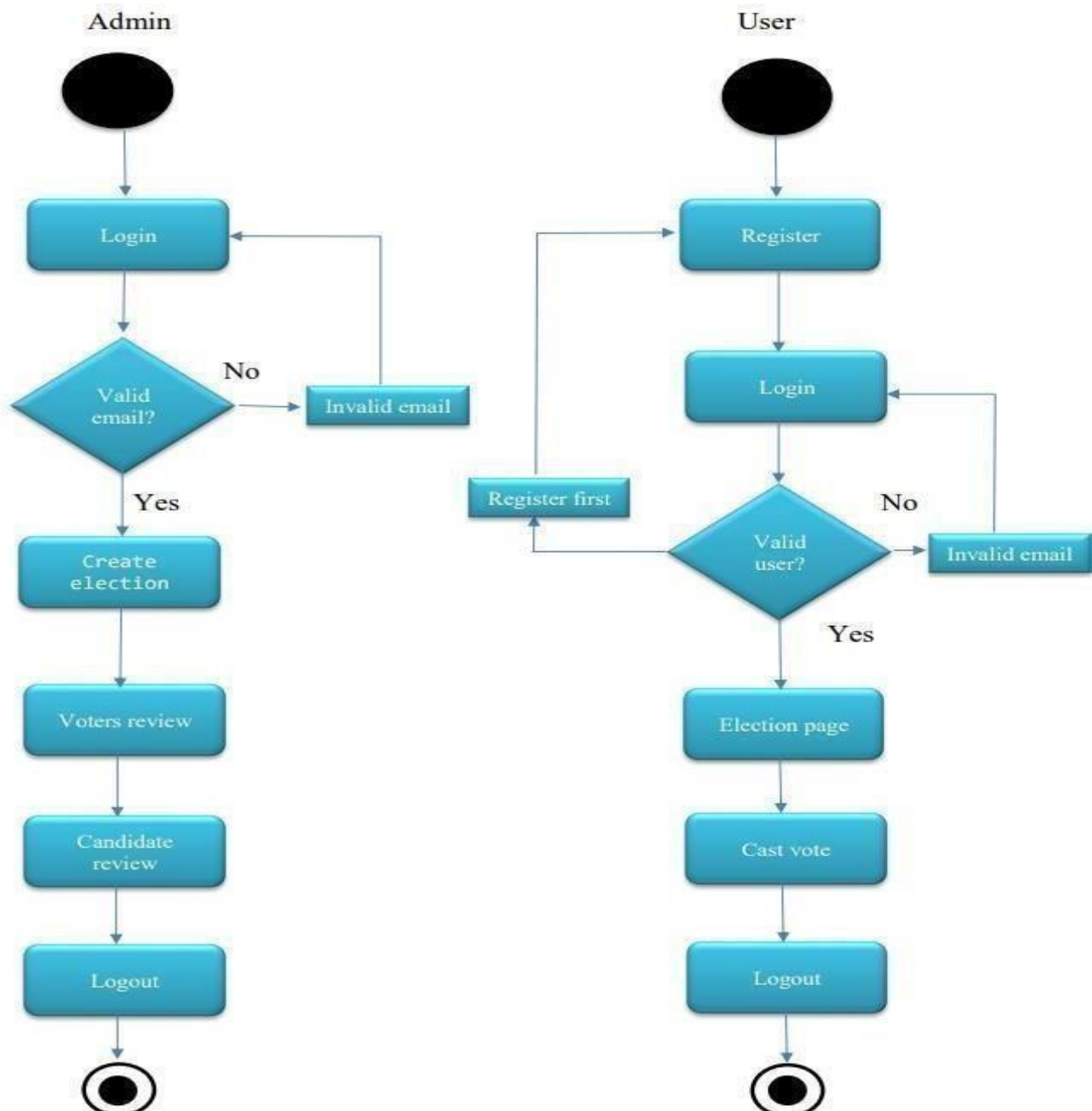


Figure 4.4: Activity Diagram



## 4.8 Sequence Diagram

Sequence diagram are an easy and intuitive way of describing the behavior of a system by viewing the interaction between the system and the environment. A sequence diagram shows an interaction arranged in a time sequence. A sequence diagram has two dimensions: vertical dimension represents time, the horizontal dimension represents the objects existence during the interaction. Basic elements:

- Vertical rectangle: Represent the object is active (method is being performed).
- Vertical dashed line: Represent the life of the object.
- X: represent the life end of an object. (Being destroyed from memory)
- Horizontal line with arrows: Messages from one object to another.

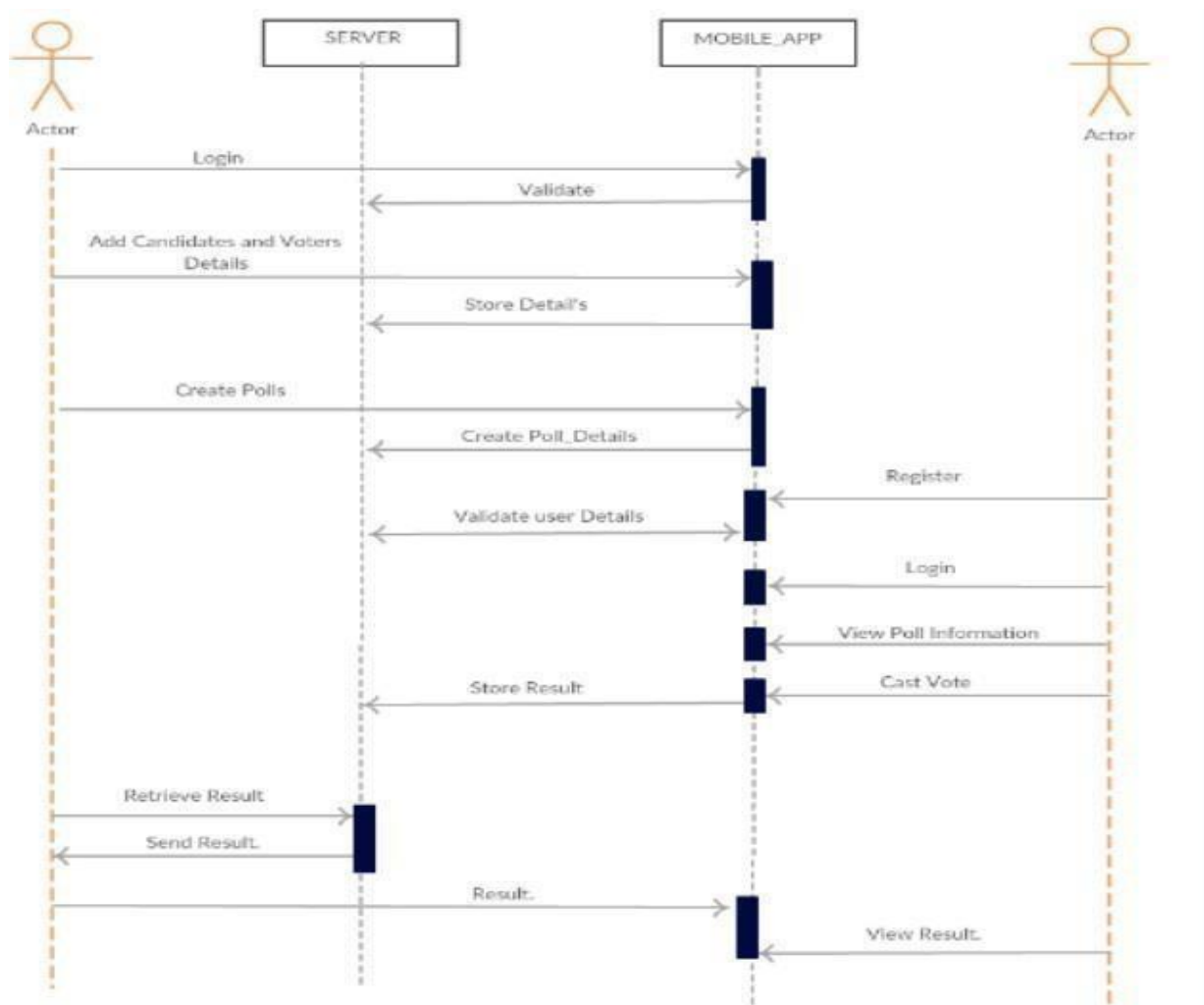


Figure 4.5: Sequence Diagram

## CHAPTER 5

# IMPLEMENTATION

### 5.1 Introduction

The implementation phase of the project is where the detailed design is actually transformed into working code. Aim of the phase is to translate the design into a best possible solution in a suitable programming language. This chapter covers the implementation aspects of the project, giving details of the programming language and development environment used. It also gives an overview of the core modules of the project with their step by step flow. The implementation stage requires the following tasks:

- Careful planning.
- Investigation of system and constraints.
- Design of methods to achieve the changeover.
- Evaluation of the changeover method.
- Correct decisions regarding selection of the platform.
- Appropriate selection of the language for application development.

### 5.2 Interfacing Android Application With Server

As we have seen ,the project must consists of two ends .One is Android application which provides interfaces for user/voter, and other is sever side which is used to store all related information of our system .First we need to create admin activity for the system ,where admin can login ,add candidate ,add voters .And as well ,we need to store the information in server ,that server is created in NETBEANS IDE .The server will be having database management system ,and all the implementation for storing the data are need to be done.AS well it will provide retrieving the existing data for client .The client is nothing but application user interface .Once after this main work is done ,we will be able to implement application for user .In which the user can register himself and login to the application after the fingerprint authentication .And the voter will caste his vote .And result calculation implement will be done.

## 5.3 Challenges And Mitigations

Despite the numerous advantages, the implementation of online voting systems is not without challenges. Addressing these challenges is essential for ensuring the successful adoption and operation of online voting platforms.

### 1. Security Concerns:

Security is the foremost concern in online voting. Cyber-attacks, data breaches, and hacking attempts pose significant risks to the integrity of the voting process. To mitigate these risks, robust security measures must be implemented, including end-to-end encryption, secure authentication protocols, and regular security audits. Employing blockchain technology can enhance security by providing an immutable record of votes that is resistant to tampering. Additionally, real-time monitoring and incident response capabilities are crucial for detecting and addressing any security threats promptly.

Voting machines are an element in a digital system, and there's no such thing as a 100% guaranteed secure digital system. However, to attack the voting machines and alter the results would present a significant set of challenges to the threat actor.

- Many different makes and models of electronic voting machines are used across the U.S. There is no standardization. The electronic voting infrastructure is made up of diverse, decentralized technologies. They're also running numerous versions and revisions of software.
- They're not connected to the Internet. Before the election, they're hardly ever turned on. This means the window of opportunity to compromise them is slim.
- Modern systems produce a verifiable paper audit that can be used to cross-reference and audit the electronic results.
- On the face of it, that sounds good. But those challenges are by no means insurmountable. And it gets worse.
- Every analysis of electronic voting systems that has been conducted has discovered basic errors that can be exploited by skilled and determined threat actors. Buffer overflows, the ability to execute remote code or install backdoors, and even backdoors pre-installed by the manufacturers have all been found.

## 2. Voter Anonymity:

Maintaining voter anonymity while ensuring the security and verifiability of votes is a critical challenge. Techniques such as blind signatures, homomorphic encryption, and zero-knowledge proofs can be employed to ensure that votes are cast and counted without compromising voter privacy. Separating the voter identification and vote casting processes can further protect anonymity, ensuring that votes cannot be traced back to individual voters.

### [1] Encryption

- **Data Encryption:** Use strong encryption methods (e.g., AES) to encrypt votes both at rest and in transit.
- **End-to-End Encryption:** Ensure that votes are encrypted on the client side before being sent to the server.

### [2] Anonymity Protocols

- **Mix Networks:** Implement mix networks or similar technologies that anonymize the source of the votes by routing them through multiple servers.
- **Zero-Knowledge Proofs:** Use cryptographic methods that allow voters to prove their vote is valid without revealing the vote itself.

### [3] Secure Authentication

- **Anonymous Authentication:** Consider using systems like blind signatures or zk-SNARKs to authenticate users without revealing their identity.
- **Multi-Factor Authentication:** Enhance security without compromising anonymity.

### [4] Decentralization

- **Blockchain Technology:** Utilize a blockchain to record votes, which can enhance transparency and security while maintaining voter anonymity.
- **Distributed Ledger:** Consider distributed systems that avoid central points of failure or control.

### [5] Auditability

- **Verifiable Voting:** Implement a method for voters to verify that their vote was counted without revealing their identity.
- **Publicly Auditable Logs:** Allow for independent audits of the vote count, ensuring that the process is transparent while maintaining anonymity.

### [6] User Interface Design

- **Clear Instructions:** Provide users with clear instructions on how their anonymity is protected.
- **User Control:** Allow users to have control over their data and the ability to opt-out of non-essential data collection.

**[7] Legal Compliance**

- **Data Protection Laws:** Ensure compliance with relevant regulations (e.g., GDPR) regarding data privacy and user anonymity.
- **Election Regulations:** Adhere to laws and standards governing online voting in your jurisdiction.

**[8] Testing and Evaluation**

- **Regular Security Audits:** Conduct regular security assessments and vulnerability testing.
- **User Feedback:** Gather feedback from users to continuously improve the system and address any concerns regarding anonymity.

**3. Digital Divide:**

The digital divide, characterized by unequal access to technology and the internet, poses a challenge to the inclusivity of online voting. To address this, efforts must be made to ensure that all eligible voters have access to the necessary technology and are equipped with the digital literacy skills required to participate in online voting. Providing alternative voting methods, such as mobile voting or assisted voting centers, can help bridge the gap for those who may not have reliable internet access or who are less comfortable with digital platforms.

**[1] Accessibility Features**

- **User-Friendly Interface:** Design an intuitive interface that accommodates users of varying tech proficiency.
- **Assistive Technologies:** Ensure compatibility with screen readers and other assistive tools for people with disabilities.

**[2] Multiplatform Access**

- **Device Compatibility:** Support a range of devices (smartphones, tablets, computers) to cater to different user preferences.
- **Offline Voting Options:** Provide alternative methods (e.g., mail-in ballots or in-person voting) for those without internet access.

**[3] Digital Literacy Programs**

- **Educational Initiatives:** Implement training programs to enhance digital literacy among voters, focusing on how to use the app effectively.

- **Community Workshops:** Organize sessions in underserved communities to teach voters about online voting.

#### [4] Infrastructure Support

- **Public Access Points:** Collaborate with libraries and community centers to provide free access to devices and internet.
- **Mobile Voting Units:** Use mobile units equipped with technology to reach remote or underserved areas.

#### [5] Language and Cultural Considerations

- **Multilingual Support:** Offer the app in multiple languages to accommodate non-English speakers.
- **Culturally Relevant Materials:** Create educational materials that resonate with diverse communities.

#### [6] Awareness Campaigns

- **Targeted Outreach:** Develop campaigns aimed at raising awareness about online voting options in marginalized communities.
- **Partnerships:** Collaborate with local organizations to spread the word and assist with registration and voting.

#### [7] Feedback Mechanisms

- **User Feedback:** Implement channels for users to report issues or suggest improvements regarding accessibility and usability.
- **Pilot Testing:** Conduct pilot tests in various communities to gather insights on barriers faced by different groups.

#### [8] Secure and Private Voting

- **Anonymity Measures:** Ensure that the app maintains voter anonymity to build trust, especially among communities concerned about privacy.

#### **4. Legal and Regulatory Framework:**

The implementation of online voting requires a robust legal and regulatory framework to ensure that the system operates within the bounds of electoral laws and regulations. This includes establishing guidelines for the development, deployment, and operation of online voting systems, as well as ensuring that these systems comply with data protection and privacy laws. Collaboration between electoral authorities, technology providers, and legal experts is essential for creating a framework that supports the secure and effective use of online voting

## CHAPTER 6

# TESTING AND RESULTS

### 6.1 Introduction

Testing is an important phase in the development life cycle of the product this was the phase where the error remaining from all the phases was detected. Hence testing performs a very critical role for quality assurance and ensuring the reliability of the software. Once the implementation is done, a test plan should be developed and run on a given set of test data. Each test has a different purpose, all work to verify that all the system elements have been properly integrated and perform allocated functions. The testing process is actually carried out to make sure that the product exactly does the same thing what is suppose to do. Testing is the final verification and validation activity within the organization itself. In the testing stage following goals are tried to achieve :-

- To affirm the quality of the project.
- To find and eliminate any residual errors from previous stages.
- To validate the software as the solution to the original problem.
- To provide operational reliabilityof the system

During testing the major activities are concentrated on the examination and modification of the source code. The test cases executed for this project are listed below. Description of the test case, steps to be followed; expected result, status and screenshots are explained with each of the test cases.

### 6.2 Testing Methodologies

There are many different types of testing methods or techniques used as part of the software testing methodology. Some of the important types of testing are:

#### 6.2.1 White Box Testing

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is purpose. It is used to test areas that cannot be reached from a black box level. Using white box testing we can derive test cases that:

- Guarantee that all independent paths within a module have been exercised at least once.
- Exercise all logical decisions on their true and false sides.



- Execute all loops at their boundaries and within their operational bounds.
- Execute internal data structure to assure their validity.

### **6.2.2 Black Box Testing**

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot see into it. The test provides inputs and responds to outputs without considering how the software works. It uncovers a different class of errors in the following categories:

- Incorrect or missing function.
- Interface errors.
- Performance errors.
- Initialization and termination errors.
- Errors in objects.

#### **Advantages:**

- The test is unbiased as the designer and the tester are independent of each other.
- The tester does not need knowledge of any specific programming languages.
- The test is done from the point of view of the user, not the designer.
- Test cases can be designed as soon as the specifications are complete.

### **6.2.3 Unit Testing**

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases. Test strategy and approach Field testing will be performed manually and functional tests will be written in detail. Test objectives

- All Components must work properly.
- Proper coordinates should be sent by the Android app to the Arduino
- The entry screen, messages and responses must not be delayed in the Android app.

### 6.3 Test Case 1

Function: void admin login()

Purpose: This function is used for admin to sign in.

Preconditions: the internet should be connected first.

Inputs: login Credentials

Expected Outputs: Should Open The admin home page.

Postconditions: Admin Can do Various Functions.

### 6.4 Test Case 2

Function: void create election()

Purpose: This function is used to create polls by adding candidates and voters.

Preconditions: Should be connected to Internet and Admin need to be logged in.

Inputs: All required information of voter and candidate will be given.

Expected Outputs: A successful message on adding candidates and voter.

Postconditions: Voters are able to register them self .

### 6.5 Test Case 3

Function: void voter reg()

Purpose: This function is used for voter to register them self.

Preconditions: The internet connection should be there.

Inputs: Voter need to give his own credential.

Expected Outputs: He will be registered for election .

Postconditions: Voter can login and do various functions .

### 6.6 Test Case 4

Function: void voter login()

Purpose: This function is used voter to sign-in to the app.

Preconditions: Internet connection should be there and he must have registered.

Inputs: He need to give username and password

Expected Outputs: He will get into The home screen.

Postconditions: He will able to see poll information and can vote

## **6.7 Test Case 5**

Function: void caste vote()

Purpose: Used to caste his vote.

Preconditions: The voter should have logged in to the app.

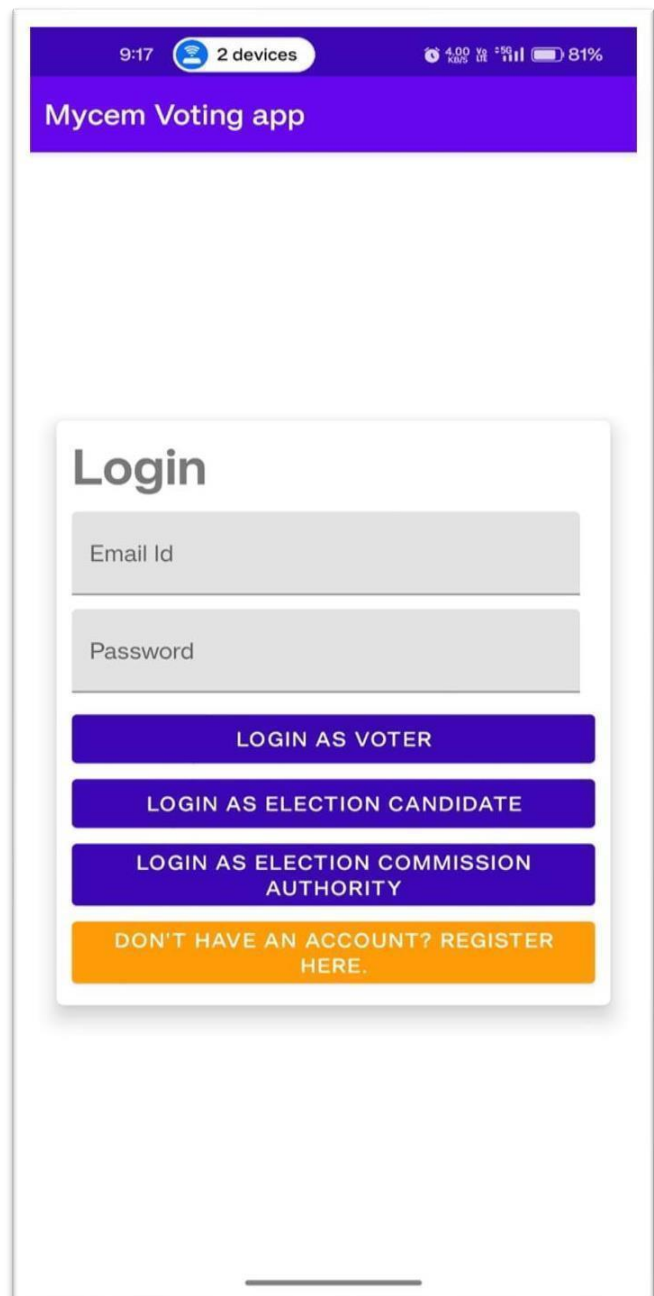
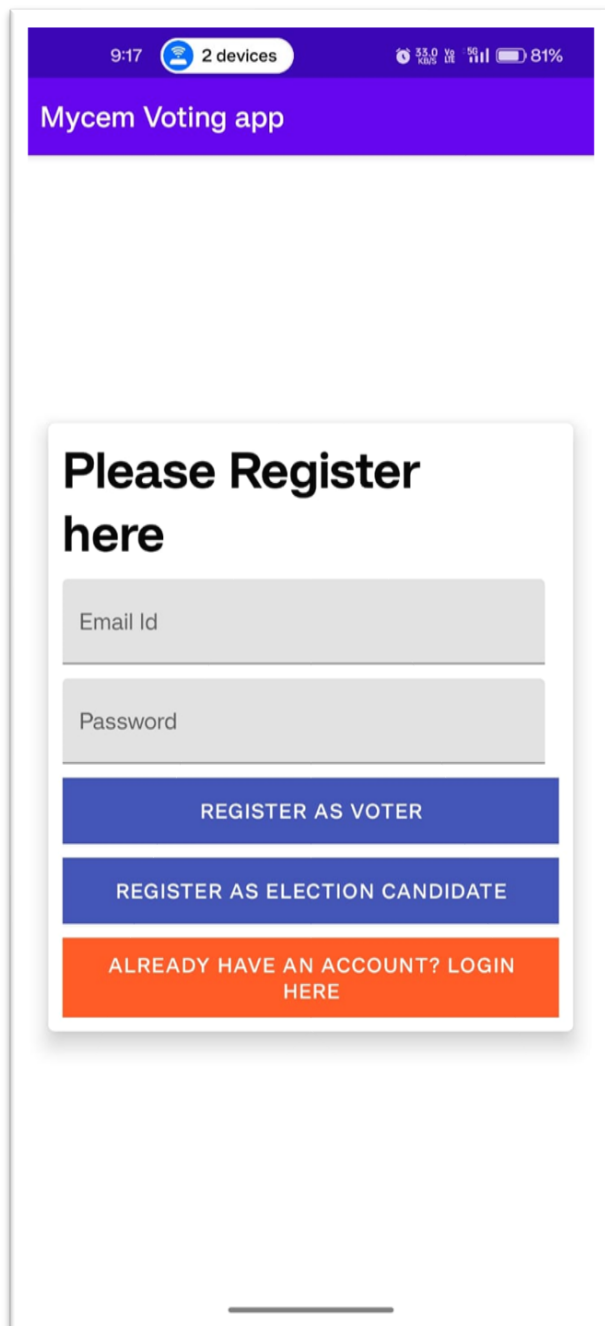
Inputs: The inputs are taken when he caste his vote.

Expected Outputs: The vote is casted successfully.

Postconditions: The system will generate final result.

# SNAPSHOT

## Images Of The Android APP



9:17 2 devices 5.00 KB/s 50% 81%

**Mycem Voting app**

Profile verified. Authorized for voting.

GO TO ELECTION PAGE GO TO EDIT PAGE

GO TO RESULTS PAGE LOGOUT

9:18 2 devices 0.30 KB/s 50% 81%

**Mycem Voting app**

**Enter Details**

Enter First Name  
deepak

Enter Last Name  
dr

☒ Male ☐ Female ☐ Others

Enter Aadhar Number  
278440988693

**Choose Date of Birth**

2000  
Sun, 24 Dec

< December 2000 >

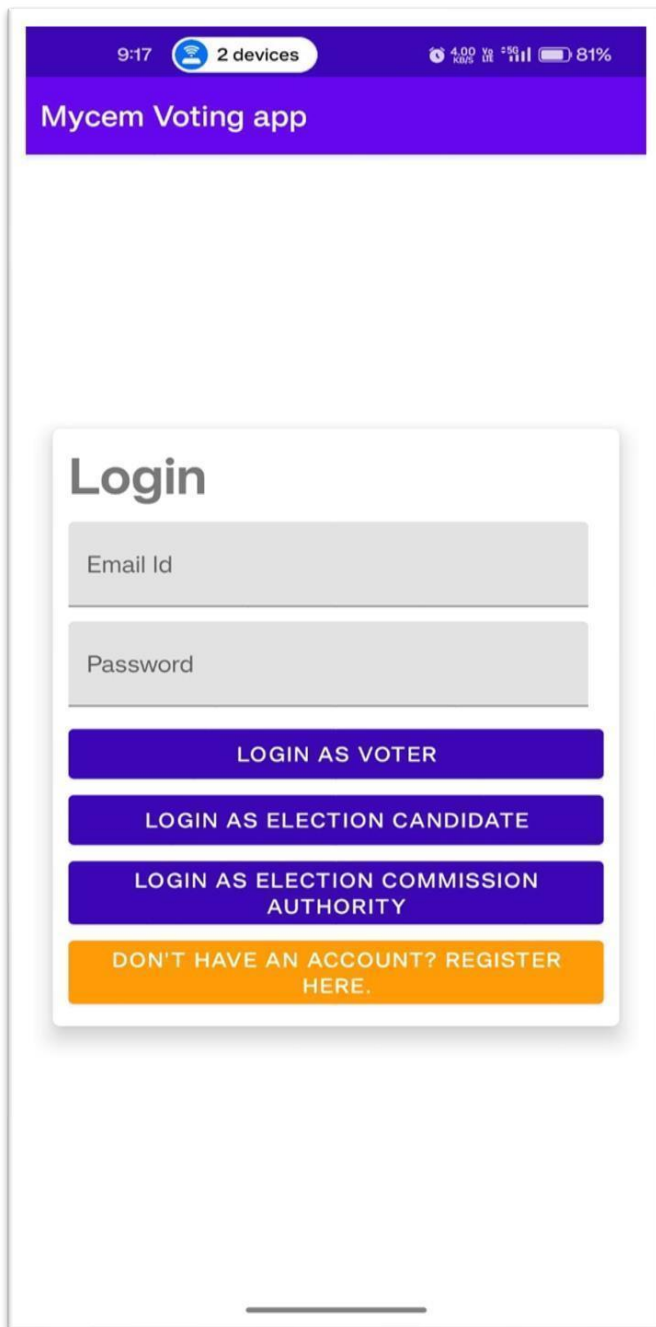
| S  | M  | T  | W  | T  | F  | S  |
|----|----|----|----|----|----|----|
|    |    |    |    |    | 1  | 2  |
| 3  | 4  | 5  | 6  | 7  | 8  | 9  |
| 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| 17 | 18 | 19 | 20 | 21 | 22 | 23 |
| 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 |    |    |    |    |    |    |

**Add Aadhar Card Image**

CHOOSE FILE SUBMIT DETAILS

GO BACK TO INFO

Snapshot.1: App Activities-1(User Activity)



9:20 2 devices 80%

Mycem Voting app

Add Election BACK TO MAIN

Election Name  
MLA

Choose Election Start Date

2024  
Tue, 9 Jul

< July 2024 >

|    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|
| S  | M  | T  | W  | T  | F  | S  |
|    | 1  | 2  | 3  | 4  | 5  | 6  |
| 7  | 8  | 9  | 10 | 11 | 12 | 13 |
| 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 |
| 28 | 29 | 30 | 31 |    |    |    |

Choose Election End Date

2024  
Tue, 9 Jul

< July 2024 >

|    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|
| S  | M  | T  | W  | T  | F  | S  |
|    | 1  | 2  | 3  | 4  | 5  | 6  |
| 7  | 8  | 9  | 10 | 11 | 12 | 13 |
| 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 |
| 28 | 29 | 30 | 31 |    |    |    |

ADD ELECTION

9:20 2 devices 80%

Mycem Voting app

First Name:  
Last Name:  
Aadhar Number:  
Gender:  
DOB:

Enter Reason in case of Declination

ACCEPTDECLINEBACK TO MAIN

Snapshot.2: App Activities-1(Admin Activity)

## CHAPTER 7

# CONCLUSION & FUTURE SCOPE

### 7.1 Conclusion

This Online Voting system will manage the Voters information by which voter can login and use his voting rights. The system will incorporate all features of Voting system. It provides the tools for maintaining voters vote to every party and it count total no. of votes of every party. There is a DATABASE which is maintained by the ELECTION COMMISSION OF INDIA in which all the names of voter with complete information is stored. In this user who is above 18 years register his/her information on the database and when he/she want to vote he/she has to login by his id and password and can vote to any party only single time. Voting detail store in database and the result is displayed by calculation. By online voting system percentage of voting is increases. It decreases the cost and time of voting process. It is very easy to use and It is vary less time consuming. It is very easy to debug.

### 7.2 Future Scope

This project was the first implementation of a system of this nature. We identify that the work done both in terms of analyzing and implementing the system is by no means complete. In this section we list the things that were either left open by this project or were opened by the analysis performed and the lessons learned during our interaction with the subject.

After discussing the benefits and features of the online voting system, it's time to think about its implementation and future scope. The implementation of the online voting system would require a significant investment in hardware, software, and internet connectivity. However, the benefits of enhanced security, a faster and automated voting process, and increased transparency justify the investment. In the future, the system could be improved by incorporating blockchain technology for added security and auditability. A more comprehensive voter database that includes biometric information, such as fingerprints or facial recognition, could enhance authentication and verification measures. While the online voting system offers many advantages over traditional paper-based voting, there are still some concerns and challenges to consider. Ensuring equal access and participation, protecting voter privacy and preventing fraud and hacking are some of the key issues that need to be addressed. In conclusion, the online voting system has enormous potential to modernize the election process in India. It is important that the results are easily accessible and that the system is implementation and future development will require a multi stakeholder approach that includes government, transparent to maintain trust in the electoral



process. Despite these challenges and concerns, addressing these issues and implementing a civil society, and the private sector. With the right strategies and investments, the online voting system could become a reliable and secure tool for democratic participation and decision-making.

### **7.3 Research Issues**

Analyze and implement different path tracking algorithms to test their effectiveness under different situations .Determine what is the correct definition of pathway quality and how to measure it .Analyze how this system can update the path to be traversed on the go.

### **7.4 Implementation Issues**

We can allow system to use separate fingerprint hardware .Implementation procedure involves proper API and hardware support .We need to have good internet connection. SDK version to be taken so that it should support fingerprint device.

## REFERENCES

- [1] Rajesh M. Ghadi1, Priyanka S. Shelar ONLINE VOTING SYSTEM ,12 | Dec-2017  
<https://www.irjet.net/archives/V4/i12/IRJETV4I12256.pdf>
- [2] Aakash1, Aashish1, Akshit1, Sarthak11 ONLINE VOTING SYSTEM, Students Dept. of Computer Science. Inderprastha Engineering College. A.P.J. Abdul Kalam Technical University , Voting 09 | March-2017 <https://www.coursehero.com/file/159339252/SSRN-id3589075pdf/>
- [3] About Android operating system [https://en.wikipedia.org/wiki/Android\\_\(operating\\_system\)](https://en.wikipedia.org/wiki/Android_(operating_system))
- [4] ONLINE VOTING SYSTEM USING CLOUD International conference on Emerging Trends in Information Technology and Engineering(ic-ETITE) 10 | Feb-2020  
[https://www.researchgate.net/publication/340972420\\_Online\\_Voting\\_System\\_using\\_Cloud](https://www.researchgate.net/publication/340972420_Online_Voting_System_using_Cloud)
- [5] [Mobile e voting system using android application for higher | PPT \(slideshare.net\)](#)